

Configuration and Reproducibility Materials

This document contains all required configuration and reproducibility materials for the Deep Learning-Based Credit Card Fraud Detection project.

1. Environment File

environment.yml

```
name: fraud-detection-env
channels:
  - defaults

dependencies:
  - python=3.10
  - pandas
  - numpy
  - matplotlib
  - seaborn
  - scikit-learn
  - tensorflow
  - jupyter
```

2. Requirements File

requirements.txt

```
pandas
numpy
matplotlib
seaborn
scikit-learn
tensorflow
jupyter
```

3. Configuration Files

.gitignore

```
data/creditcard.csv
*.zip
```

```
.ipynb_checkpoints/  
__pycache__/
```

Purpose:

- Prevents large dataset files from being uploaded to GitHub
 - Ignores temporary notebook checkpoint files
 - Keeps repository clean and reproducible
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4. Data Instructions

Dataset Source

Dataset used:

Credit Card Fraud Detection Dataset (Kaggle)

<https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>

Dataset Setup Instructions

1. Download the dataset from Kaggle.
2. Extract the ZIP file.
3. Copy `creditcard.csv` into the following folder:

`data/creditcard.csv`

4. Ensure the final dataset path is:
`data/creditcard.csv`
-

5. Retraining Instructions

To reproduce the project results:

Step 1: Install Dependencies

```
pip install -r requirements.txt
```

Step 2: Start Jupyter Notebook

```
jupyter notebook
```

Step 3: Open Notebook

Open the notebook file:

```
credit_card_fraud_detection.ipynb
```

Step 4: Run All Cells

Run all notebook cells sequentially to:

- Load dataset
 - Preprocess data
 - Train deep learning model
 - Evaluate model
 - Generate visualizations
-

6. Training Configuration

The following training configuration was used:

Parameter	Value
Random State	42
Train-Test Split	80/20
Batch Size	2048
Epochs	10
Optimizer	Adam
Learning Rate	0.001
Loss Function	Binary Crossentropy

7. Model Reproducibility Notes

- Class weighting was applied to handle severe class imbalance.
- Fraud detection performance was evaluated using:
 - Recall
 - Precision
 - F1-score
 - ROC-AUC
- Results may vary slightly depending on hardware and TensorFlow version.

8. Saved Weights / Model Files

Saved model weights are not included in the repository due to repository size considerations.

The model can be fully retrained using the provided notebook and dataset instructions.

9. Repository Structure

```
credit-card-fraud-detection/  
├── README.md  
├── requirements.txt  
├── environment.yml  
├── .gitignore  
├── credit_card_fraud_detection.ipynb  
├── report.pdf  
  
└── data/  
    └── README.md
```

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